MENGYU LI

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Google Scholar: https://scholar.google.com/citations?user=hPWnWK0AAAAJ&hl=en

APPOINTMENTS

Department of Statistics and Data Science, Tsinghua University (THU), Beijing, China

- Postdoctoral Fellow Aug. 2025–present
- Advisor: Prof. Jun S. Liu

EDUCATION

Institute of Statistics and Big Data, Renmin University of China (RUC), Beijing, China

- Ph.D. in Statistics Sep. 2020–Jun. 2025
- Supervisor: Assoc. Prof. Cheng Meng

Department of Statistics, The Chinese University of Hong Kong (CUHK), Hong Kong, China

- Exchange Student Nov. 2023–May 2024
- Advisor: Prof. Xiaodan Fan

School of Statistics, Beijing Normal University (BNU), Beijing, China

- Bachelor in Statistics Sep. 2016–Jun. 2020
- Thesis advisor: Prof. Junlong Zhao

RESEARCH INTERESTS

Optimal transport; Statistical computing; Generative modeling; Statistical machine learning; AI for science

PUBLICATIONS

1. **Mengyu Li**, Qianqian Qu, Jun S. Liu* (2026). "BridgeTwist: Twisting the Schrödinger bridges for training-free conditional sampling." NeurIPS 2026.
2. Junyi Lin, **Mengyu Li**†, Cheng Meng*, Yongdao Zhou* (2026). "Sparsification subsampling for partial least squares regression." Journal of Computational and Graphical Statistics. In press.
3. Dunyao Xue, **Mengyu Li***, Cheng Meng, Jingyi Zhang (2026). "Core-elements subsampling for alternating least squares." Journal of Computational and Graphical Statistics. In press.
4. Xiaxue Ouyang, Hao Zheng, Haoxian Liang, Jingyi Zhang, Yixuan Qiu, Cheng Meng, **Mengyu Li*** (2026). "Sparsification techniques for large-scale optimal transport problems." Wiley Interdisciplinary Reviews: Computational Statistics. 18(1), e70056.
5. **Mengyu Li**, Cheng Meng*, Xiaodan Fan (2026). "Iterative optimal transport for multimodal image registration." Pattern Recognition. 172(PD).

∗ Corresponding author; † Joint first author; ‡ Alphabetical order

6. Junlie Huang, Xinlai Kang, Qiannan Huang, **Mengyu Li**, Cheng Meng* (2025). "Efficient approximation of leverage scores in two-dimensional autoregressive models with application to image anomaly detection." Journal of Computational and Graphical Statistics. 35(1), 89–100.
7. Tao Wang, **Mengyu Li**†, Geduo Zeng, Cheng Meng*, Qiong Zhang* (2025). "Gaussian herding across pens: An optimal transport perspective on global Gaussian reduction for 3DGS." NeurIPS 2025. Spotlight (Top 3%).
8. **Mengyu Li**, Bencong Zhu†, Cheng Meng*, Xiaodan Fan* (2025). "Double optimal transport for differential gene regulatory network inference with unpaired samples." Bioinformatics. 41(8), btaf352.
9. Yukuan Hu, **Mengyu Li**‡, Xin Liu, Cheng Meng (2025). "Sampling-based methods for multi-block optimization problems over transport polytopes." Mathematics of Computation. 94(353), 1281–1322.
10. **Mengyu Li**, Jun Yu†, Tao Li, Cheng Meng* (2024). "Core-elements for large-scale least squares estimation." Statistics and Computing, 34(6), 1–16.
    Student Paper Award in JSM2022, Sections on Statistical Computing and Graphics
11. **Mengyu Li**, Jingyi Zhang, Cheng Meng* (2024). "Nonparametric additive models for billion observations." Journal of Computational and Graphical Statistics, 33(4), 1397–1412.
12. Xinlai Kang, **Mengyu Li**, Xuqiang Chen, Fangyu Li, Cheng Meng* (2023). "A Hausdorff regression paradigm for interval privacy." IEEE Signal Processing Letters, 31, 146–150.
13. **Mengyu Li**, Jun Yu†, Tao Li, Cheng Meng* (2023). "Importance sparsification for Sinkhorn algorithm." Journal of Machine Learning Research, 24(247), 1–44.
    Student Paper Award in JSM2023, Statistics in Imaging Section
14. **Mengyu Li**, Jun Yu†, Hongteng Xu, Cheng Meng* (2023). "Efficient approximation of Gromov-Wasserstein distance using importance sparsification." Journal of Computational and Graphical Statistics, 32(4), 1512–1523.
15. **Mengyu Li**, Junlong Zhao* (2022). "Communication-efficient distributed linear discriminant analysis for binary classification." Statistica Sinica, 32, 1343–1361.

Working Papers

1. **Mengyu Li**, Meng Cheng*, Jun S. Liu*. "Free-support Wasserstein resampling in sequential Monte Carlo." Minor revision in *Journal of the American Statistical Association*.
2. Xiaxue Ouyang, **Mengyu Li**, Jun Yu, Cheng Meng*. "A more efficient method for large-sample model-free feature screening via multi-armed bandits." Revision in *Statistics and Computing*.
3. Pengyu Li, **Mengyu Li**†, Jianzhu Ma, Jun S. Liu. "Steer, select, and stratify: Training-free sampling for protein motif scaffolding." Submitted.
4. Xinyue Yang, **Mengyu Li***, Jun S. Liu*. "Generative distributional regression via joint and conditional distribution matching." Submitted.
5. **Mengyu Li**, Shunxing Yan†, Jun S. Liu*. "Localized Gaussian mixture filtering with Gaussian-space optimal transport resampling." Ready to submit.

FUNDING & AWARDS

Funding

- Youth Science Foundation Project (Category C), National Natural Science Foundation of China (¥300,000) 2027–2029
- General Project, China Postdoctoral Science Foundation (¥80,000) 2025–2027
- Shuimu Tsinghua Scholar Program 2025–2027
- Young Elite Scientists Sponsorship Program - Doctoral Student Special Program, China Association for Science and Technology (¥40,000) 2024–2025
- Outstanding Innovative Talents Cultivation Funded Program (¥70,000), RUC 2021–2024
- Joint Ph.D. Program (¥50,000), RUC 2023–2024

Awards

- **Wu Yuzhang Scholarship**, RUC 2025
  RUC's highest honor for students; 10 winners (among which 4 Ph.D. students) per year
- **Outstanding Poster Award** in the 22nd Annual Meeting of China Society for Industrial and Applied Mathematics (CSIAM 2024) 2024
- **Student Paper Award** in Joint Statistical Meetings (JSM), Statistics in Imaging Section, American Statistical Association (ASA) 2023
  5 winners per year; first winner trained in Chinese Mainland
- **Student Paper Award** in JSM, Sections on Statistical Computing and Graphics, ASA 2022
  4 winners per year; first winner trained in Chinese Mainland
- Beijing Municipal Outstanding Graduate 2025
- National Scholarship, Ministry of Education of the P.R. China 2023
- Jingdong Special Scholarship, RUC 2023
  20 winners per year

ACADEMIC EXPERIENCE

Research Assistant

- Department of Statistics, CUHK Nov. 2023–May 2024
  Developed novel algorithms for multi-modal image registration and gene regulatory network reconstruction based on optimal transport theory.

Teaching Assistant

- Data Mining, RUC Spring 2021, 2024
  My duties included lecturing exercise lessons, homework preparation and grading, etc.

Talks & Poster Presentations

- The 3rd Joint Conference on Statistics and Data Science in China, Hangzhou, China Jul. 2025
- 2025 ICSA China Conference, Zhuhai, China Jun. 2025
- RUC Mathematics Time, Renmin University of China, Beijing, China Apr. 2025
- School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China Mar. 2025
- PKU Applied Math Lunch Seminar, Peking University, Beijing, China Dec. 2024
- The 8th CSIAM Student Forum in CSIAM 2024, Nanjing, China Oct. 2024

- Student Academic Seminar for the 87th Anniversary of RUC, Beijing, China Oct. 2024
- Capital Statistics and Data Science High-Level Forum, Beijing, China Jul. 2023

Referee for Journals and Conferences

- Journal of Machine Learning Research
- Journal of Computational and Graphical Statistics
- Annals of Applied Statistics
- Statistica Sinica
- Communications in Statistics - Simulation and Computation
- SIAM Journal on Imaging Sciences
- Bioinformatics Advances
- Statistical Applications in Genetics and Molecular Biology
- NeurIPS, ICLR

PROFESSIONAL SKILLS

Programming

- Python, R, MATLAB, C++, Linux shell

Language

- Chinese Mandarin, English